



Group 4: Process Pioneers

Process Improvement Plan - Online Ordering System

Velvet Espresso Café

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Velvet Espresso Café

High-quality coffee and pastries, a welcoming neighborhood experience, open 24 hours.



Mission, Vision & Operations

Mission

Quality coffee & food with fast, friendly service — building customer loyalty.

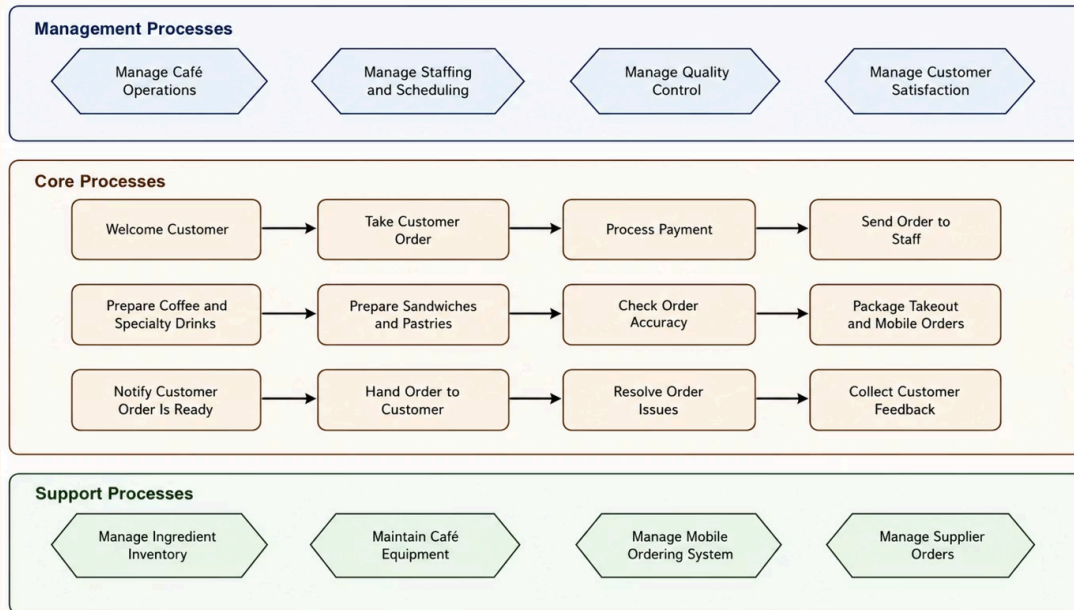
Vision

Preferred neighborhood café through exceptional service and innovative technology.

Core Operations

- Order & payment processing
- Inventory & supplier management
- Staff scheduling & quality assurance
- Marketing & financial reporting

Process Landscape Model

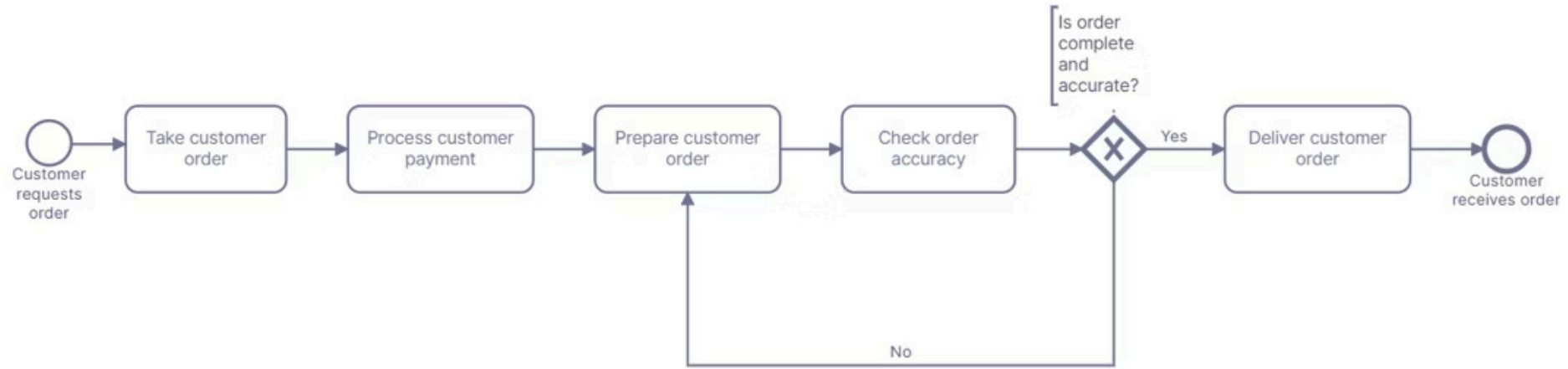


Target for Improvement

Customer Order Processing — the primary value-creating process. Directly impacts speed, accuracy, satisfaction, and efficiency.

☐ Connects payments, preparation, inventory, and quality control.

As-Is Process Model



Single decision point creates a rework loop — the primary source of delays during peak hours.

Discovery Phase: Data

13min

Avg Wait Time

Peak-hour observation of 30 orders

13%

Order Corrections

4 of 30 orders needed remakes

10%

Customers Left

3 of 30 abandoned the line

60%

Custom Orders

18 of 30 included customizations

Stakeholder Personas



Customer

A college student who visits before class and wants fast service. Their main concerns are long lines, unclear wait times, and incorrect customizations.



Cashier

Takes orders, answers questions, enters customizations, and processes payments. Their main concern is making mistakes while rushing during busy hours.



Barista

Prepares drinks and needs clear instructions. Their main concerns are unclear customizations, sudden rushes, and remaking incorrect orders.



Shift Manager

Manages employees, inventory, and customer complaints. Their main concerns are unpredictable demand, employee stress, product shortages, and lost sales.

AI-Driven Improvements

AI Voice Ordering

Hands-free ordering. **Risk:** complex requests → comment-box fallback.

Predictive Wait Times

AI estimates pickup time using order volume & staff speed. **Risk:** new staff learning curve → resolves over time.

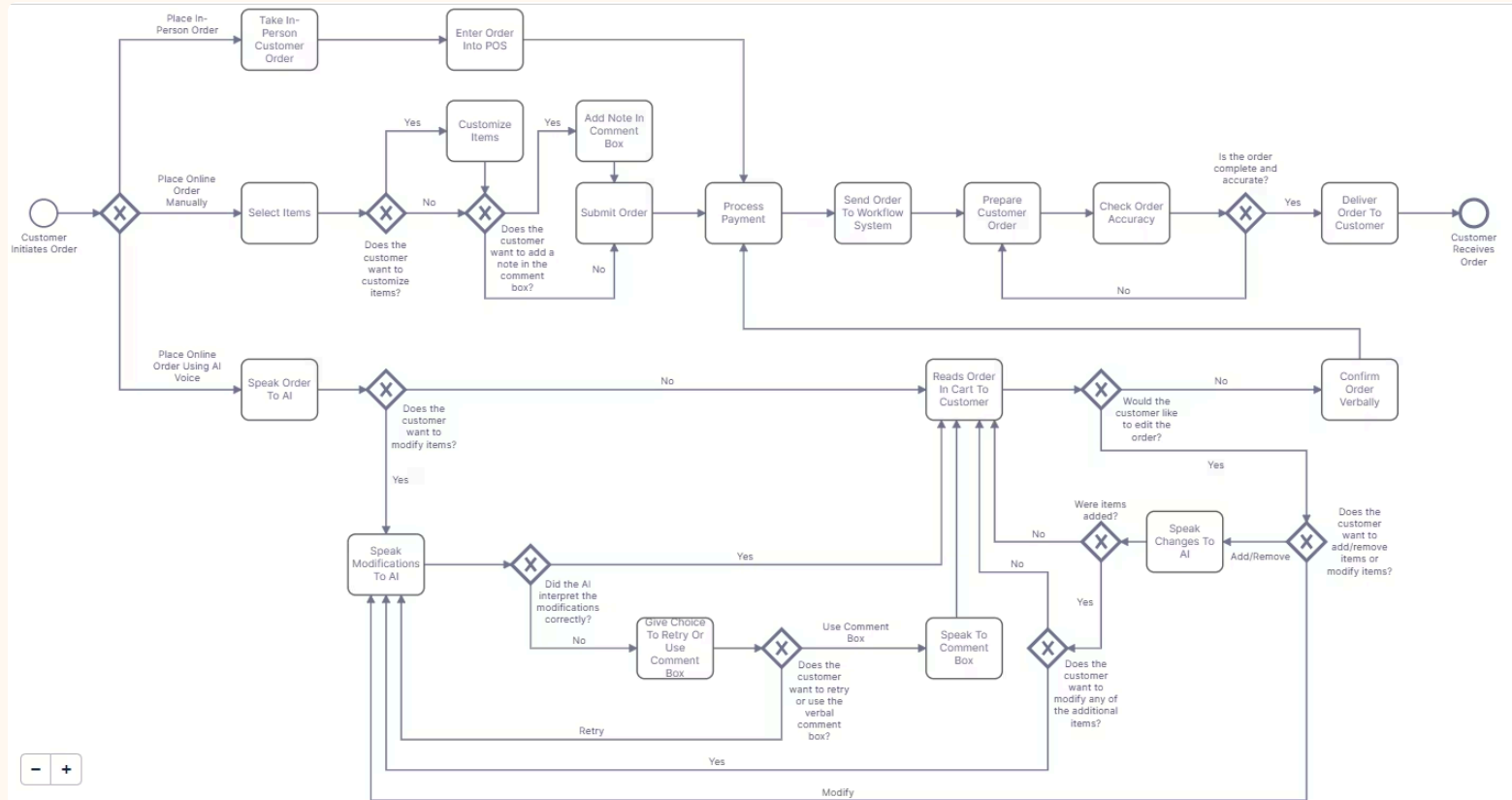
AI Chatbot

Answers customer queries, freeing staff. **Risk:** knowledge gaps → continuous Q&A training.

 Inspired by Stord (AI delivery predictions) & Gojek (AI voice assistant) — Google Cloud gen AI use cases.



To-Be Model & KPIs



Target KPIs

 Processing Time

↓ 35–40% reduction

✓ Order Accuracy

≥ 90% via AI voice + comment box

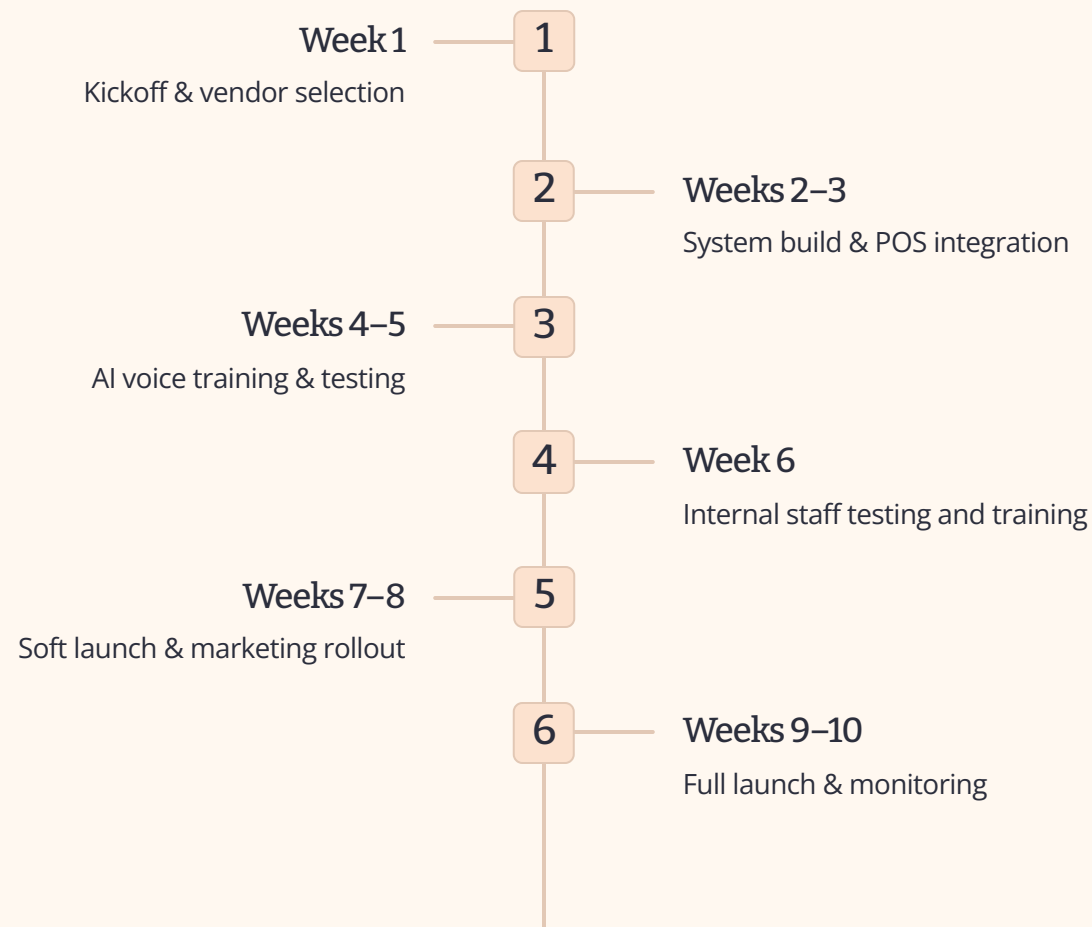
★ CSAT Score

≥ 4.5 / 5

KPI Dashboard



Implementation Timeline



📌 Key stakeholders: IT Team, Internal IT Team, Operations/Project Manager, AI Vendor, Shift Supervisors, Frontline Staff, Marketing Team, Customer Experience Team.

Monitoring & Continuous Improvement



Performance Tracking

- Order volume & adoption rate
- Avg order time & AI error rate
- CSAT, NPS & repeat usage



Feedback Sources

- Qualtrics surveys post-purchase
- POS audits & remake logs



Improvement Cycle

1. Collect Feedback
2. Analyze Trends
3. Prioritize Fixes
4. Implement Updates
5. Re-Survey

Qualtrics Survey

Thank you for taking the time to complete this survey. The purpose of this survey is to gather feedback on the effectiveness of our newly implemented online ordering system. Your input helps us understand what is working well and what may need improvement. We value your response and review all feedback to guide updates and enhancements.

1. Compared to ordering in person, how does the online ordering experience feel?

- ☐ Much Better
- ☐ Somewhat Better
- ☐ About The Same
- ☐ Somewhat Worse
- ☐ Much Worse

2. What device did you use to place your order?

- ☐ Smartphone
- ☐ Tablet
- ☐ Laptop
- ☐ Desktop Computer

3. Which features did you use during your order?

- ☐ Menu Browsing
- ☐ Customizing Items
- ☐ Saving A Favorite Order
- ☐ AI Voice Ordering

4. Please rate your agreement with the following statement:
"The website was easy to navigate."

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

5. Please rate your agreement with the following statement:
"Placing my order was quick and efficient."

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

6. Please rate your agreement with the following statement:
"I am likely to use the online ordering system again."

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

7. Overall, how satisfied are you with your experience using our online ordering system?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

8. How likely are you to recommend our online ordering system to a friend or coworker?

Not At All Likely Extremely Likely

0	1	2	3	4	5	6	7	8	9	10
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9. Did any part of the online ordering experience stand out to you in a positive way? If so, please describe it.

10. What improvements or new features would you like to see added to our online ordering experience?

Link:

https://fiu.qualtrics.com/jfe/form/SV_dcLOzxAeGSPWa9M

Thank You!

Questions?

We appreciate your time and interest in our process improvement plan for Velvet Espresso Café.

